

# Computer Networks Lecture 02: OSI Model

## 1 Introduction to the OSI Model

When two computers communicate with each other, several operations must be performed before the data reaches its destination. The data is created by an application, formatted, prepared for transmission, divided into smaller units when necessary, provided with addresses, organized into frames, and finally transmitted as physical signals.

To organize these operations, the Open Systems Interconnection (OSI) model divides network communication into seven layers.

The seven layers, starting from the sender's application and moving toward the communication medium, are:

1. Application Layer
2. Presentation Layer
3. Session Layer
4. Transport Layer
5. Network Layer
6. Data Link Layer
7. Physical Layer

Each layer performs specific functions and provides services to the layer above it. At the sender, data moves from the Application Layer down to the Physical Layer. At the receiver, the process moves upward in the reverse order.

## 2 Layer 7 - Application Layer

### 2.1 Definition

The Application Layer is the highest layer of the OSI model. It provides network services and an interface through which user applications access network communication.

It is the layer closest to the user. It contains protocols that allow applications to exchange information over a network.

Examples of Application Layer protocols include HTTP, HTTPS, FTP, DNS, SMTP, and Telnet.

## **2.2 Explanation**

When a user performs a network activity, the application needs a protocol to communicate with another system.

For example, when a user opens a website, the browser uses HTTP or HTTPS to communicate with a web server. When a user sends an email, an email application may use SMTP to send the message.

The Application Layer prepares the application-level information that will be passed to the lower layers.

## **2.3 Actual Working**

1. Provides network services to user applications.
2. Creates or receives application-level data.
3. Uses protocols such as HTTP, FTP, DNS, and SMTP.

## **2.4 Example**

A user enters `www.example.com` into a web browser. The browser uses HTTP or HTTPS to prepare a request for the web server. The request is then passed to the lower layers.

# **3 Layer 6 - Presentation Layer**

## **3.1 Definition**

The Presentation Layer is the sixth layer of the OSI model. It is responsible for representing and formatting data in a form that the receiving system can understand.

It can also provide encryption, decryption, compression, and data conversion.

## **3.2 Explanation**

Data created by an application may need to be converted into a suitable representation before transmission. Different systems may use different formats or encodings.

The Presentation Layer provides a conceptual place for handling these differences. It may convert data into a suitable format, compress it to reduce its size, or encrypt it to protect its confidentiality.

At the receiving side, the data may be decrypted, decompressed, or converted into the format required by the application.

In modern networks, these functions are often implemented by application software or security protocols rather than by a separate Presentation Layer.

### **3.3 Actual Working**

1. Formats and converts data into a suitable representation.
2. Encrypts or decrypts data when required.
3. Compresses or decompresses data when required.

### **3.4 Example**

A user sends a document over a secure connection. The data may be represented in a suitable format and encrypted before transmission. The receiving system decrypts the information when it has the appropriate key.

## **4 Layer 5 - Session Layer**

### **4.1 Definition**

The Session Layer is the fifth layer of the OSI model. It is responsible for establishing, managing, and terminating communication sessions between applications.

A session is a logical communication relationship between two applications or systems.

### **4.2 Explanation**

Some forms of communication require a session to be established before information can be exchanged. The session must remain active while the communication continues and may need to be closed when the work is complete.

The Session Layer describes the functions used to manage this communication relationship. It can also support dialog control and synchronization points during a long communication process.

The Session Layer is concerned with managing a session, while the Transport Layer provides transport services for moving data between applications.

In modern Internet systems, session-related functions are often handled by application protocols and software.

### **4.3 Actual Working**

1. Establishes a logical session between applications.
2. Manages the session during communication.
3. Terminates the session when communication is finished.

## 4.4 Example

A user logs in to a remote service and performs several operations. The service maintains a logical session for the user's activity and ends it when the user logs out or the session expires.

# 5 Layer 4 - Transport Layer

## 5.1 Definition

The Transport Layer is the fourth layer of the OSI model. It provides communication between applications or processes running on different computers.

It can divide large amounts of data into smaller units and provide services such as reliability, flow control, and error recovery.

## 5.2 Explanation

A large file or message may need to be divided into smaller pieces for transmission. This process is called segmentation.

The receiving Transport Layer can use the information in these units to put the data back into the correct order and reconstruct the original information.

The Transport Layer also uses port numbers to help deliver data to the correct application. Different transport protocols provide different services.

TCP provides reliable and ordered delivery. It uses mechanisms such as acknowledgements and retransmission. UDP provides a simpler transport service without the same built-in reliability mechanisms.

## 5.3 Actual Working

1. Divides application data into smaller transport units when required.
2. Provides end-to-end communication between applications or processes.
3. Provides services such as reliability, flow control, and error recovery when supported by the protocol.

## 5.4 Example

A user downloads a large file using TCP. The data is transmitted in smaller units. TCP helps ensure that missing data is retransmitted and that the data is delivered in the correct order.

## **6 Layer 3 - Network Layer**

### **6.1 Definition**

The Network Layer is the third layer of the OSI model. It is responsible for logical addressing and delivering packets across interconnected networks.

It determines how packets can travel from the source network toward the destination network.

### **6.2 Explanation**

When a computer communicates with a device on another network, the data needs logical addressing information.

The Network Layer uses IP addresses to identify the source and destination. Routers examine network-layer information and forward packets toward their destinations.

The process of selecting a suitable path through networks is called routing. The actual movement of packets to the next network device is called forwarding.

A packet may pass through several routers before reaching its destination.

### **6.3 Actual Working**

1. Adds logical addressing information, such as source and destination IP addresses.
2. Determines or uses routing information to select a path toward the destination.
3. Forwards packets between networks.

### **6.4 Example**

A computer in Pakistan sends a request to a web server in another country. The packet contains the source and destination IP addresses. Routers forward it through different networks until it reaches the destination.

## **7 Layer 2 - Data Link Layer**

### **7.1 Definition**

The Data Link Layer is the second layer of the OSI model. It organizes packets into frames and provides communication over a local network link.

It also handles local addressing, media access, and error detection.

## 7.2 Explanation

The Network Layer produces packets. The Data Link Layer places each packet inside a frame suitable for the local network technology.

Frames contain control information that helps devices identify and process the transmitted data. On Ethernet and Wi-Fi networks, MAC addresses are used for local delivery.

The Data Link Layer can also detect certain errors that occur during transmission. For example, an Ethernet frame contains a field used to check whether the frame has been corrupted.

Error detection does not mean that every error is automatically corrected. The response depends on the technology and protocol being used.

The Data Link Layer does not generally provide data confidentiality. Encryption is used when information must be protected from being read by unauthorized parties.

## 7.3 Actual Working

1. Encapsulates network-layer packets into frames.
2. Uses local addressing, such as MAC addresses, for link-level delivery.
3. Detects certain transmission errors in frames.

## 7.4 Example

A laptop sends data to a Wi-Fi router. The Data Link Layer creates a Wi-Fi frame containing the packet and the required local network information. The frame is then passed to the Physical Layer.

# 8 Layer 1 - Physical Layer

## 8.1 Definition

The Physical Layer is the lowest layer of the OSI model. It is responsible for transmitting raw bits through a communication medium.

It deals with the physical signals used to carry information.

## 8.2 Explanation

After the data has passed through the higher layers, it must be transmitted through a physical medium.

The Physical Layer represents bits using signals. The type of signal depends on the technology being used.

Ethernet cables use electrical signals, fiber optic cables use light signals, and wireless networks use radio signals.

At the receiving side, the Physical Layer detects the incoming signals and converts them into bits for the Data Link Layer.

### **8.3 Actual Working**

1. Converts bits into physical signals for transmission.
2. Sends signals through the communication medium.
3. Receives signals and converts them back into bits.

### **8.4 Example**

When a computer sends data through an Ethernet cable, the Physical Layer represents the bits using electrical signals. In a Wi-Fi network, radio signals carry the information through the air.

## **9 Complete Flow from Sender to Receiver**

Consider a user sending a file to another computer.

At the sender, the process begins at the Application Layer. The application creates the data and passes it downward.

The Presentation Layer formats the data and may encrypt or compress it. The Session Layer manages the communication session. The Transport Layer divides the data into smaller units when required and provides end-to-end transport services.

The Network Layer adds logical addressing information and supports routing. The Data Link Layer places packets into frames for local delivery. Finally, the Physical Layer transmits the bits as physical signals.

At the receiving computer, the process occurs in reverse. The Physical Layer receives the signals, and the Data Link Layer processes the frames. The Network Layer processes the packets, and the Transport Layer delivers the data to the appropriate application process.

The upper layers then perform any required session management, data conversion, decryption, or decompression. Finally, the Application Layer makes the information available to the receiving application.

## **10 Important Note**

The OSI model is a reference model used to understand and describe network communication. Modern Internet protocols do not always implement seven separate layers exactly as described.

For example, session management, data formatting, and encryption may be handled by application protocols or software libraries. Similarly, the exact functions of each layer depend on the technology being used.

The important concept is that network communication involves different responsibilities, and the OSI model organizes these responsibilities into seven layers.

## 11 References

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